



HenCheck

FIND. FLAG. FOCUS.

DELAWARE DMV

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Vendor: DelDot

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With HenCheck





With HenCheck



THE SOLUTION

TODAY, WITHOUT HENCHECK

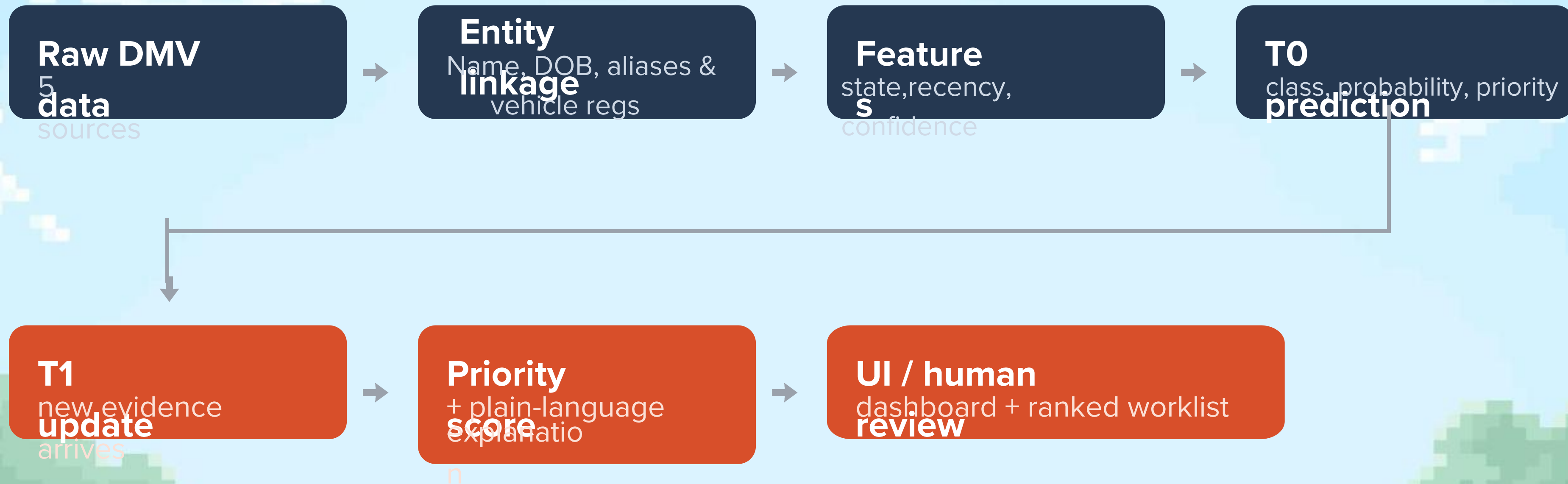
- Evidence is spread across multiple data sources
- Address, credential, title, work, and external signals must be considered together
- Old out-of-state history can be difficult to distinguish from current conflicting evidence
- New evidence at T1 can change the review picture
- Reviewing fragmented evidence case-by-case can be time-consuming and inconsistent

VS

WITH HENCHECK

- Records linked across sources into one candidate view
- Evidence compared across address, credential, title, work, and external sources
- Case scored across 3 review outcomes + confidence + priority
- T0 and T1 evidence changes tracked explicitly
- Key evidence and conflicts surfaced for the reviewer
- Reviewer receives a ranked, explainable worklist while retaining the final decision

End to end workflow

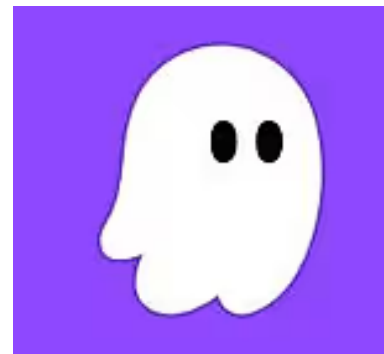


COMPUTE PLATFORM & LEARNING



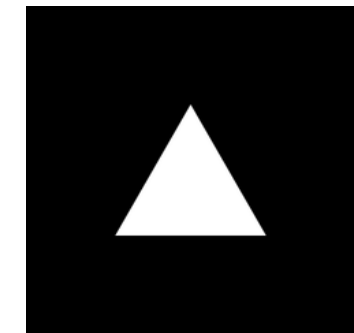
Google Cloud Platform

- Deployed the full app on Cloud Run
- Cloud Build → Artifact Registry → Cloud Run
- Tried BigQuery



AWS Kiro

- Main coding agent
- Caught edge cases we hadn't thought to test
- Tried deploying using AWS via Kiro; couldn't get it working



Vercel AI

- Vercel hosted the dashboard as a fast fallback
- Auto-detected the framework and built it

Azure

- Not explored, time constraint

Limitations

DATA & LINKAGE

- No shared key between evidence and candidates
- Only ~56% match on exact name + DOB
- Uncertain links abstain rather than guess

MODELING

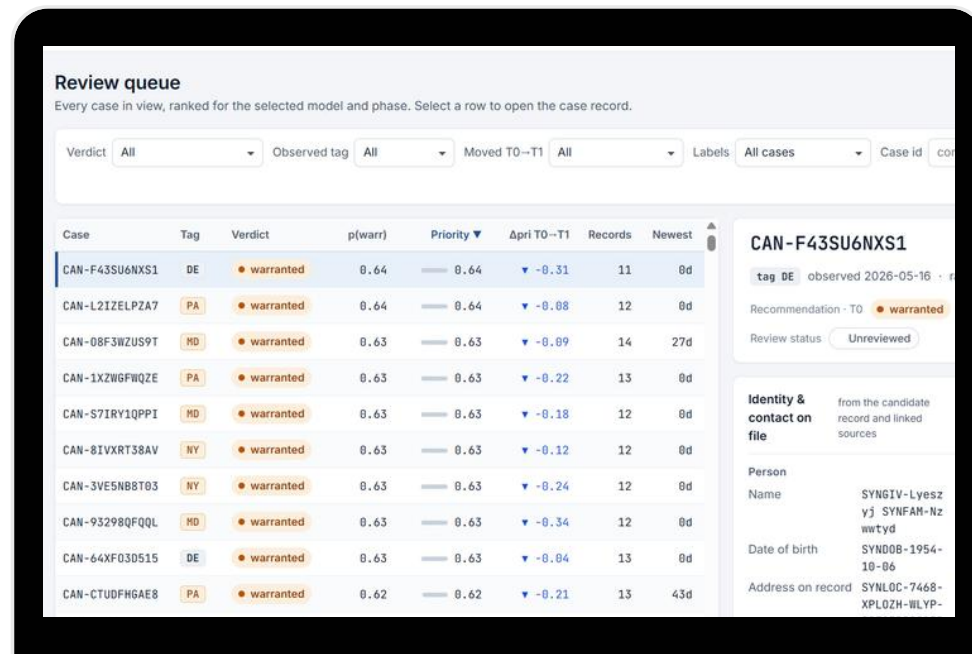
- 300 labeled cases out of 12,000 — 2.5% of the population
- Log loss 1.09 · Macro F1 0.51 · Calibration error 0.09

DEPLOYMENT

- Local first build, not production-hardened
- Never tested against real DMV records — synthetic data only
- Local first build

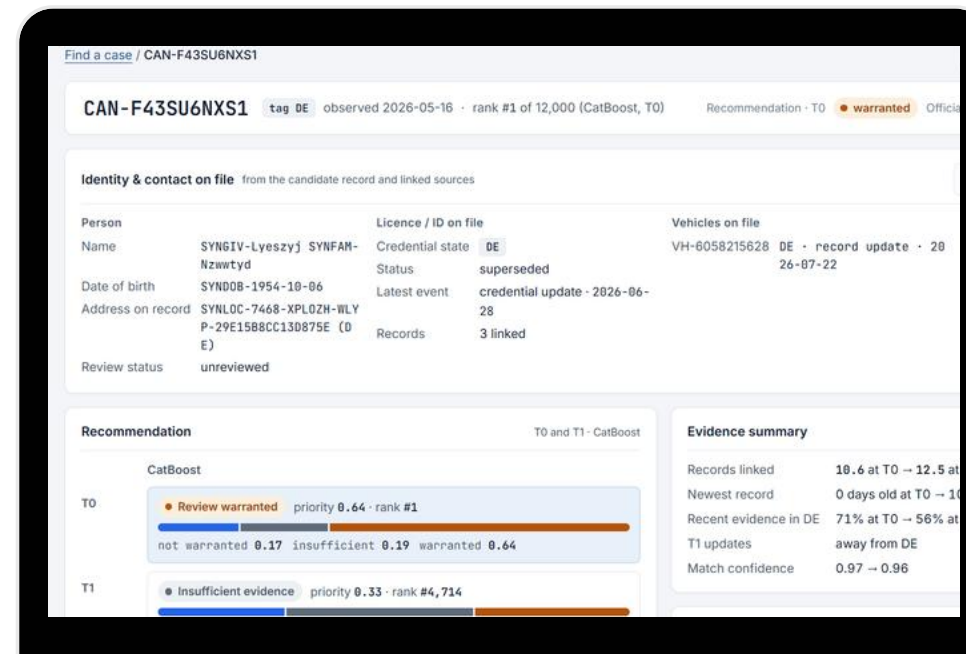
What we Built: HenCheck

Review Queue



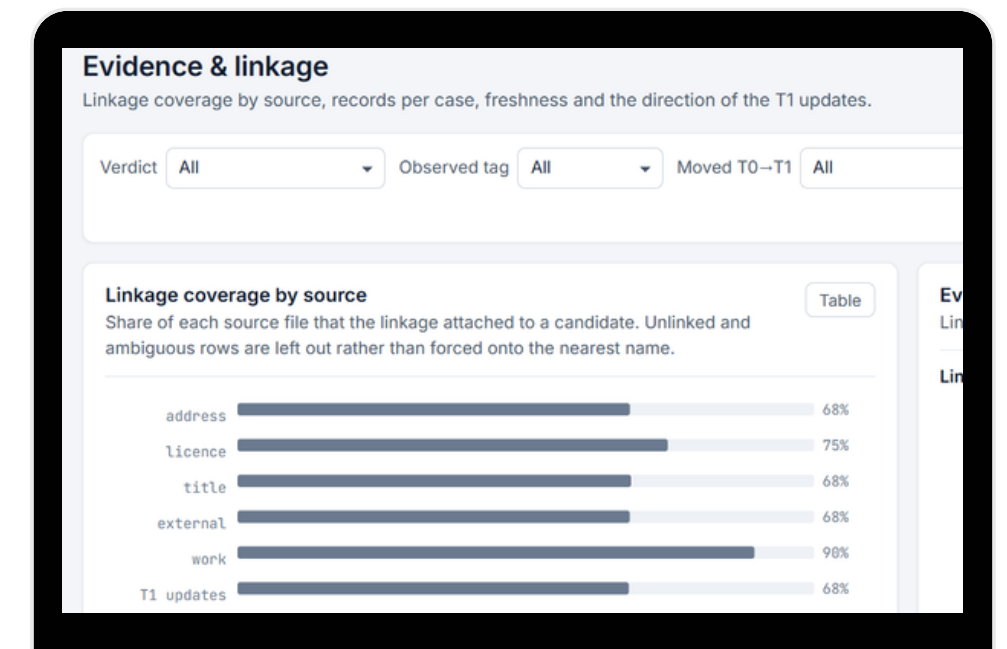
Prioritizes all cases so reviewers can quickly focus on the records that need attention.

Case Investigation



Shows prediction confidence, priority, T0→T1 changes, and why a case was classified.

Evidence Records



Brings address, credential, title, work, and external records together in one place for review.

demo here: <https://first-state-tag-review.vercel.app/>

Delaware Impact

Delaware DMV staff benefit from faster, more focused case review

Reduces manual cross-verification across multiple datasets

Helps prioritize cases that are more likely to need attention

Saves staff time and improves review consistency

Can help reduce missed registrations and potential revenue loss

For a pilot: connect agency data, validate results with DMV reviewers, and deploy the dashboard securely in the cloud

Even a modest recovery rate on unregistered out-of-state vehicles represents real, currently uncollected Delaware registration revenue.

RESPONSIBLE AI & DATA



Data: fully synthetic, no real residents, no real PII



Handles corrupted identity data, abstains when unsure instead of guessing



Reports confidence, not just a label, insufficient evidence is a real answer

**What's different: we don't just link records,
we prove our matches are trustworthy and
prove our model isn't cheating, with
automated tests, not just claims.**

**Even a modest recovery rate on unregistered out-of-
state vehicles represents *real*, currently uncollected
Delaware registration revenue.**

Takeaway:

- **Problem:** Manual cross-checking is slow and hard to prioritize.
- **Solution:** One workspace connects evidence and ranks cases.
- **Result:** Reviewers quickly see what needs attention and why.
- **Impact:** Saves time, improves consistency, and helps protect Delaware revenue.

*Turn thousands of disconnected records into a **clear, prioritized path** for human review.*

**This concludes our presentation,
any questions?**

